

Question	Answer	Marks
9(a)(i)	$eV = hc / \lambda$	C1
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (84 \times 10^3 \times 1.60 \times 10^{-19})$ $= 1.5 \times 10^{-11} \text{ m}$	A1
9(a)(ii)	either (some) kinetic energy (of electrons) is converted to thermal energy at target or some X-rays are absorbed by the target so its temperature increases	B1
	(tungsten) has higher melting point so does not melt quickly / easily	B1

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9(b)	$I = I_0 \exp(-\mu t)$	C1
	$0.13 = [\exp(-3.0x)] \times [\exp(-0.22x)]$ $= \exp(-3.22x)$	C1
	$x = 0.63 \text{ cm}$	A1
9(c)(i)	product of density and speed	M1
	speed of ultrasound in medium	A1
9(c)(ii)	$I_R / I_0 = (7.8 - 1.7)^2 / (7.8 + 1.7)^2$ $= 0.41$ fraction transmitted = $1.00 - 0.41 = 0.59$	C1
	percentage transmitted = 59%	A1
9(c)(iii)	more than one <u>boundary</u> so more reflections	B1
	some ultrasound is attenuated in matter	B1

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